An Explicit Class-Number-One Cutoff from Recent Real-Zero Gaps

A response to Tao's tenth-discriminant challenge

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Abstract

Assuming the explicit real-zero estimates in Banks (for $q \geq 10^{10}$) and Lu–Zaman–Zhao (for $q \leq 10^{10}$), we prove that a negative fundamental discriminant D with imaginary-quadratic class number one satisfies $|D| < 50{,}000$. A short Dirichlet-hyperbola inequality, positivity of the coefficients of $\zeta(s)L(s,\chi_D)$ and their square-index contribution provide an elementary zero-gap-to-$L(1)$ bridge. An exact count of reduced primitive positive binary quadratic forms below the cutoff gives the nine familiar fields. The cited zero-free theorems are treated as external inputs rather than re-proved here. This note is a quantitative extraction, not a new Siegel-zero theorem, and does not improve previously known numerical class-number cutoffs.

Precise source dependencies. We use Banks [3] for the explicit gap $(1-\beta)\log q \geq 3/1000$ for real zeros of primitive real Dirichlet characters with conductor $q \geq 10^{10}$. For $q \leq 10^{10}$ we use the real-axis exclusion $L(\sigma,\chi) \neq 0$ at $\sigma \geq 1 - 1/(5\log q)$ from Lu–Zaman–Zhao [4, Theorem 1.1]. These are external inputs, not results newly proved in this note. Standard classical analytic continuation, the imaginary-quadratic class-number formula and proper binary-form/ideal-class correspondence are also used. No class-number bridge from Kimura [6] or Heegner's classification enters the deduction.

1 Result and quantitative context

Proposition 1 (Effective cutoff from the two cited real-zero bounds)**.** *Let $D < 0$ be a negative fundamental discriminant and let $h(D)$ be the class number of $\mathbb{Q}(\sqrt{D})$. Under the two Banks/Lu–Zaman–Zhao inputs above,*

$$\boxed{h(D) = 1 \quad \Longrightarrow \quad |D| < 50{,}000.}$$

An independent exact enumeration of reduced primitive binary quadratic forms for $3 \leq |D| < 50{,}000$ gives class number one exactly for

$$|D| \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}.$$

Thus the same assumptions, plus the elementary finite check, reproduce the classical complete nine-field list without importing Heegner's classification theorem. This is not *a claim of a new theorem beyond existing published work.*

Tao's challenge [2], motivated in part by the Tao–Green discussion of interpolation determinants [1], asks for numerical bounds extracted from different approaches to Siegel-zero exclusion. The present proof uses the explicit zero-free estimates of [3, 4]; it does not effectivize the Tao–Green prime-nonresidue estimate directly.

A significant prior-art comparison is the unconditional result of Bennett, Martin, O'Bryant and Rechnitzer [5, Proposition 1.10]: for a primitive quadratic character of conductor $q > 6677$, they proved $L(1,\chi) > 12/\sqrt{q}$. In the imaginary quadratic case this implies $h(D) > 12/\pi > 3$ whenever $|D| > 6677$. Thus the cutoff 50,000 below is *not* an improvement over previously known numerical class-number bounds. Its purpose is to exhibit a short explicit deduction from the specified 2026 Banks/Lu–Zaman–Zhao inputs, as a comparison of proof techniques.

2 An elementary class-number bridge

Let χ be a nonprincipal primitive real character modulo q. Put

$$a(n) = (1 * \chi)(n) = \sum_{d|n} \chi(d), \qquad A(t) = \sum_{n \le t} a(n), \qquad L_1 = L(1, \chi).$$

By the Euler factors, $a(n) \ge 0$ for all n, $a(1) = 1$, and

$$a(m^2) \ge 1 \quad \text{for every } m \ge 1. \tag{1}$$

Lemma 2 (Explicit Dirichlet-hyperbola estimate)**.** *For every real $t \ge 1$,*

$$\boxed{|A(t) - tL_1| \le 7q\sqrt{t}.}$$

Proof. Put $y = \lfloor \sqrt{t} \rfloor$ and $S_\chi(v) = \sum_{n \le v} \chi(n)$. Since χ is nonprincipal and periodic modulo q, complete periods sum to zero; hence $|S_\chi(v)| \le q$ for every real $v \ge 0$. Dirichlet's test and partial summation give $\left|L_1 - \sum_{d \le y} \chi(d)/d\right| \le 2q/y$. The exact hyperbola identity is

$$A(t) = \sum_{d \le y} \chi(d) \left\lfloor \frac{t}{d} \right\rfloor + \sum_{m \le y} S_\chi(t/m) - y S_\chi(y).$$

The first sum differs from $t \sum_{d \le y} \chi(d)/d$ by at most y; the other two terms contribute at most $2qy$. Therefore

$$|A(t) - tL_1| \le y + \frac{2qt}{y} + 2qy \le (1 + 6q)\sqrt{t} \le 7q\sqrt{t},$$

using $\frac{1}{2}\sqrt{t} \le y \le \sqrt{t}$ and $q \ge 1$. □

Lemma 3 (Zero gap to $L(1)$)**.** *Suppose $q \ge 50{,}000$, $0 < c \le 1/5$, and $L(s, \chi)$ has no real zero in $(1 - c/\log q, 1)$. Then*

$$\boxed{\sqrt{q}\, L(1, \chi) \;\ge\; \frac{c}{\log q} \left(\frac{13}{8} e^{-3c} \sqrt{q} - 28 \right).} \tag{2}$$

Proof. Set $\delta = c/\log q$, $\sigma = 1 - \delta$ and $X = q^3$. Here $3/4 < \sigma < 1$. Since $L_1 > 0$ for quadratic characters and there are no zeros in $(\sigma, 1)$, continuity gives $L(\sigma, \chi) \ge 0$. Also $\zeta(\sigma) < 0$, so $\zeta(\sigma) L(\sigma, \chi) \le 0$.

With $E(t) = A(t) - L_1 t$, partial summation for $\Re(s) > 1$ and analytic continuation to $\Re(s) > 1/2$, $s \ne 1$, give the exact identity

$$\zeta(s) L(s, \chi) = \sum_{n \le X} \frac{a(n)}{n^s} + \frac{L_1 X^{1-s}}{s - 1} - E(X) X^{-s} + s \int_X^\infty E(t) t^{-s-1}\, dt.$$

The last two terms have total absolute value at most

$$7qX^{1/2-\sigma} \left(1 + \frac{\sigma}{\sigma - 1/2} \right) \le 28 q X^{1/2-\sigma} = \frac{28 e^{3c}}{\sqrt{q}}.$$

By (1), $\sigma < 1$ and $X \ge 50^2$,

$$\sum_{n \le X} \frac{a(n)}{n^\sigma} \ge \sum_{m=1}^{50} \frac{1}{m^{2\sigma}} \ge \sum_{m=1}^{50} \frac{1}{m^2} > \frac{13}{8},$$

where the last inequality is an exact finite rational sum. It follows that

$$0 \ge \zeta(\sigma) L(\sigma, \chi) \ge \frac{13}{8} - \frac{L_1 X^\delta}{\delta} - \frac{28 e^{3c}}{\sqrt{q}}.$$

As $X^\delta = q^{3c/\log q} = e^{3c}$, rearrangement proves (2). □

3 Two exact numerical regimes

For fundamental $D = -q$ with $q > 4$, the analytic class-number formula reads

$$h(D) = \frac{\sqrt{q}}{\pi} L(1, \chi_D).$$

For every fixed $c > 0$, the function

$$f_c(q) = \frac{\frac{13}{8} e^{-3c} \sqrt{q} - 28}{\log q}$$

is strictly increasing for $q > e^2$, since its derivative has numerator $\frac{13}{8} e^{-3c} \sqrt{q} (\frac{1}{2} \log q - 1) + 28 > 0$.

Range $50{,}000 \le q \le 10^{10}$. Use the Lu–Zaman–Zhao zero exclusion [4, Theorem 1.1], with $c = 1/5$. Exact rational checks establish

$$e^{-3/5} > \frac{100}{183}, \quad \sqrt{50{,}000} > \frac{447}{2}, \quad \log(50{,}000) < \frac{1083}{100}, \quad \pi < \frac{22}{7}.$$

The exponential bound uses a degree-four Taylor polynomial with a dominated geometric tail, and the logarithm bound follows from $e^{10} > 22{,}000$ and $e^{83/100} > 25/11$, both from finite positive Taylor sums. Therefore (2) gives the *strict* bound, uniformly throughout this range,

$$h(D) > \frac{(1/5)\,[(13/8)(100/183)(447/2) - 28]}{(22/7)(1083/100)} = \frac{1455755}{1453386} > 1.$$

Range $q \ge 10^{10}$. Apply Banks's large-conductor result [3] with $c = 3/1000$. Now $e^{-9/1000} > 991/1000$, $\sqrt{q} \ge 100{,}000$ and $\log(10^{10}) < 24$ (indeed $\log 10 < 12/5$, from a finite Taylor sum). By the same monotonicity,

$$h(D) > \frac{(3/1000)\,[(13/8)(991/1000)(100{,}000) - 28]}{(22/7) \cdot 24} = \frac{2254133}{352000} > 1.$$

These inequalities prove the claimed cutoff $|D| < 50{,}000$.

4 Finite exact enumeration and independence

For $q = |D| < 50{,}000$, enumerate *negative fundamental* discriminants without class-number tables. They are precisely $D = -q$ with either $D \equiv 1 \pmod 4$ squarefree or $D = 4m$, where $m \equiv 2, 3 \pmod 4$ is squarefree. For each such D, count the positive primitive reduced integral binary quadratic forms

$$(a, b, c): \quad b^2 - 4ac = D, \quad |b| \le a \le c, \quad \gcd(a, b, c) = 1,$$

with $b \ge 0$ when $|b| = a$ or $a = c$. It suffices to check $1 \le a \le \lfloor \sqrt{q/3} \rfloor$. By the classical correspondence with proper ideal classes, this count equals $h(D)$.

Two independent source-code implementations were used. Enumerator A loops through all fundamental discriminants and then through (a, b); enumerator B loops through reduced triples (a, b, c) and groups by discriminant. Both find exactly 15,195 negative fundamental discriminants in the range and agree *on every single class-number count*. Their unique class-number-one values are

$$\boxed{3, 4, 7, 8, 11, 19, 43, 67, 163.}$$

These correspond to $d = 3, 1, 7, 2, 11, 19, 43, 67, 163$ in Tao's $\mathbb{Q}(\sqrt{-d})$ squarefree-radicand notation.

5 Scope and AI assistance

This note does *not* make Tao–Green's unquantified prime-nonresidue estimate explicit. Instead, it uses Banks's explicit determinant-derived real-zero gap and a separate finite zero computation of Lu–Zaman–Zhao. Kimura's recent work [6] gives considerably broader class-number bounds and formal verification with a different argument; the elementary bridge here is presented for clarity and quantitative comparison, with no claim of first discovery of its constituent techniques.

AI assistance. This human-directed investigation used GPT-6 (Extra High) to assist with mathematical derivation, exact computation and exposition. The deduction and finite checks were subsequently independently reviewed by a separate fresh-context GPT-Pro model. That review accepted the cited Banks and Lu–Zaman–Zhao zero-free results as inputs; it was an AI review, not human peer review. Attribution for those analytic results belongs to their original authors and the underlying OpenAI mathematical work.

Notation. The bound is for the absolute field discriminant $|D|$, which equals the conductor of the primitive quadratic character. Tao's nine squarefree radicands $d = 1, 2, 3, 7, 11, 19, 43, 67, 163$ correspond to $|D| = 4, 8, 3, 7, 11, 19, 43, 67, 163$.

References

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